Week 4 Day 5 — Monitoring Checklist

Final model: tuned Gradient Boosting, threshold 0.40 | Aqsa Saqib

What to monitor and when to alert

What to watch	How	Normal value	Alert when	Check
Input problems	Rows that fail validation; missing values; unseen categories	Missing about 6% (workclass, occupation)	More than 2% of rows fail, or missing values double	Daily
Data drift	PSI for age, hours, capital gain, education, marital status, occupation	PSI below 0.1	PSI above 0.2 for any key feature	Weekly
Model score	Average predicted chance and share predicted >50K	About 25% predicted >50K	Below 18% or above 32%	Weekly
Label distribution	Real share of >50K when results come back	About 24%	Changes by more than 5 points	Monthly
Performance	F1, precision and recall on new labelled data	F1 about 0.72	F1 below 0.68 or precision below 0.65	Monthly
Fairness	F1 and recall by sex and race	Sex gap about 0.01	A gap grows by more than 0.05	Monthly
Service	Prediction time and error count	Well under 100 ms	Slower than 100 ms, or errors above 1%	Daily

Retraining cadence

- **Regular:** retrain every 6 months on the newest labelled data.
- **Early:** retrain when a drift or performance alert lasts 2 weeks.
- **Before switching:** test the new model on a hold-out set, compare F1 and group scores with the current model, and switch only if it is at least as good.
- **Keep a record:** model version, training date, data period, threshold and test scores for every model.

When an alert fires

- **Input problems:** check the data source first. The model may be fine, but the data pipeline may be broken.
- **Drift or score change:** look at which feature moved, check if the customer group changed, and plan retraining.
- **Performance drop:** pause automatic decisions if F1 falls below 0.65, and use the old selection method until fixed.
- **Fairness gap:** review the affected group with the team before the model keeps running.

Note: PSI (population stability index) compares how a feature is spread today with how it was spread in the training data. Below 0.1 = no change, 0.1 to 0.2 = small change, above 0.2 = big change.